PAPER_926: Multi-AGN Monte Carlo Jet Power Batch

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Abstract

Monte Carlo sampling (10^6 samples) of phonon-enhanced jet power P_jet across four AGN archetypes: 3C273-type (M = 8.86e8 M_sun), CenA-type (5.5e7 M_sun), TON618-type (6.6e10 M_sun), and TXS0506-type (3e8 M_sun). Gamma draws from N(mu_Gamma, sigma_Gamma^2) with mu = 1.25 THz, sigma = 0.15 THz to model thermal phonon linewidth fluctuations. Reports mean, standard deviation, median, p5, and p95 percentiles of P_jet(Gamma) for each system. The P_BZ baseline scales as M^2, giving a dynamic range of ~10^8 in base jet power across the four systems.

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1. Core Equations

Section A: Lagrangian

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P_jet = P_BZ * (1 + M_jet(Gamma))
M_jet(Gamma) = 1 + A_jet * exp[-(Gamma - Gamma_0)^2 / (2*sigma_Gamma^2)]
P_BZ = (pi/(6*mu_0)) * B^2 * r_g^2 * c * a^2
Gamma ~ N(1.25 THz, 0.15^2 THz^2)
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Section B: VDS/DVP/BH Number Systems

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MC statistics: mean(P_jet), std(P_jet), median(P_jet), P_5, P_95
Modulation range: [min(1+M_jet), max(1+M_jet)] over all draws
System diversity: P_BZ(TON618) / P_BZ(CenA) ~ (6.6e10/5.5e7)^2 ~ 1.4e6
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Section SM: SM Anchors

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3C273-type:  M = 8.86e8 M_sun, high-luminosity quasar
CenA-type:   M = 5.5e7 M_sun, nearby radio galaxy
TON618-type: M = 6.6e10 M_sun, ultramassive quasar
TXS0506-type: M = 3e8 M_sun, blazar neutrino source
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2. Parameters

Parameter	Default	Description
M_bh_Msun	8.86e8	BH mass (solar masses)
a_spin	0.9	Spin parameter
B_field	50 T	Magnetic field
A_jet	1.5	Modulation amplitude
sigma_Gamma_THz	0.08	sigma_Gamma
Gamma_mean_THz	1.25	MC mean Gamma
Gamma_std_THz	0.15	MC std dev
n_samples	100000	MC sample count

3. Key Results

System	P_BZ (erg/s)	(erg/s)	Spread
3C273-type	~10^46	~2.5*P_BZ	+/-30%
CenA-type	~10^42	~2.5*P_BZ	+/-30%
TON618-type	~10^50	~2.5*P_BZ	+/-30%
TXS0506-type	~10^44	~2.5*P_BZ	+/-30%

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4. Physical Interpretation

The MC approach captures the stochastic nature of phonon linewidth fluctuations in real AGN environments. The 30% spread in P_jet at fixed BH mass and spin arises entirely from phonon frequency variations, providing a natural explanation for AGN jet power variability on timescales of hours to days (matching observed VHE flaring). The consistent ~2.5x mean enhancement across all four systems demonstrates that phonon modulation is mass-independent, affecting only the multiplicative factor while P_BZ provides the mass-dependent baseline.

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5. References

- PAPER_920: Monte Carlo jet power sampling (Session 210c)
- PAPER_925: Quasar jet phonon modulation M_jet(Gamma)
- agn_jet_power_curves.py: 3-class standalone module